

MASM22/FMSN30/FMSN40/FMSN30F SPRING 2026

LAB 1: LEAD IN MOSS — CHANGE OVER TIME

Uses the techniques from Lecture 1 and 2

Deadline: Thursday 2 April 2026

Dedicated help sessions:

Wednesday 25 March at 10.15–12.00 and 13.15–15.00.

Examination: Mozquizto Lab 1:A+B+C

Perform the tasks by copying relevant R-code from Lecture 1 and 2, rewrite and run it while answering the questions in the accompanying three Mozquizto-tests at `quizms.maths.lu.se`. These tests will also provide you with the information marked `[mzq]` below.

1 Lab 1: Problem description

In order to assess the concentration levels of poisonous metals in the environment, Naturvårdsverket (Swedish Environmental Protection Agency) measures the levels of a number of different metals in moss every five years. Since the largest sources of emission have disappeared (been banned) during the later decades, the levels in nature are declining over time.

We have access to the measurements from a handfull of Swedish regions in the file `Pb_all.rda`. Download the data file and save it in the subfolder `Data` in your RStudio project folder. Start the RStudio project and create an R-script file, e.g., `lab1.R`, and save it in the subfolder `R` in your RStudio project folder. Write and run the necessary R-code in the script file.

Start by adding a header comment and initiate the package collection `tidyverse`. Alternatively you can use only the packages `ggplot2` (for plotting) and `dplyr` (for data manipulation). Then load the data and look at the summary:

```
#Lab1####  
library(tidyverse)  
load("Data/Pb_all.rda")  
summary(Pb_all)
```

Start the Mozquizto-test *Lab 1.A: Linear model* in order to find out which region you should analyse. Extract the data for your region (replace `[mzq]` as appropriate) and look at the summary and the first few lines:

```
Pb_myregion <- filter(Pb_all, region == "[mzq]")  
summary(Pb_myregion)  
head(Pb_myregion)
```

The variable `Pb` is the measured concentration level of lead (Pb) in mg/kg dry weight, that is, the amount of mg lead per kg dried moss. We are interested in how fast the lead concentration decreases.

1.A Linear model

1.A(a). Plot Pb against year:

```
##1.A Linear model####
###1.A(a)####
ggplot(Pb_myregion, aes(x = year, y = Pb)) + geom_point()
```

Does the relationship look linear? Should it be linear?

- 1.A(b). The measurements started in 1975 and it would be more reasonable to use that as a starting point instead of year = 0. Replot the data using $x = \text{year} - 1975$ instead of $x = \text{year}$.
- 1.A(c). Ignore the non-linear appearance and fit a linear regression model $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ where $Y = \text{Pb}$ and $x = \text{I}(\text{year} - 1975)$. The $\text{I}()$ is necessary in order to use a function, “-” = “subtraction”, that R could confuse with “-” = “leave this variable out”. Calculate the β -estimates, their standard errors and 95 % confidence intervals.
- 1.A(d). What is the average concentration of lead in 1975, according to this model?
- 1.A(e). How does the concentration change over time, on average, according to this model?
- 1.A(f). Use the model to calculate a 95 % confidence interval for the expected average lead concentration in the year [mzq]. Does it seem reasonable?
- 1.A(g). Use the model to calculate a 95 % prediction interval for observations of the lead concentration in the year [mzq]. Does it seem reasonable?
- 1.A(h). Plot the data together with the fitted line, a 95 % confidence interval for the fitted line and a 95 % prediction interval for new observations. Does it look reasonable?
- 1.A(i). Calculate the residuals and plot them against the predicted values. As visual aides, add a horizontal reference line at 0 and a moving average, + `geom_smooth()`. Any problems here?
- 1.A(j). Make a Q-Q-plot and a histogram for the residuals. Problems?

1.B Log-transformation

Continue with Mozquizto-test *Lab 1.B: Log-transformation*.

- 1.B(a). The concentration of lead, when there are no new additions, should decrease at a constant rate, e.g., 10 % per year. What type of model would this indicate and what type of transformations would be natural for Y and/or x ?
- 1.B(b). Plot $y = \log(\text{Pb})$ against $x = \text{year} - 1975$. Does this seem like a linear relationship?
- 1.B(c). Fit this log-transformed linear model and calculate the β -estimates, their standard errors and 95 % confidence intervals.
- 1.B(d). What is the average log-concentration of lead in 1975, according to this model?

- 1.B(e). How does the log-concentration change over time, on average, according to this model?
- 1.B(f). Use the model to calculate a 95 % confidence interval for the expected average log-lead concentration in the year `[mzq]`.
- 1.B(g). Use the model to calculate a 95 % prediction interval for the logarithm of an observation of the lead concentration in the year `[mzq]`.
- 1.B(h). Plot the log-transformed data together with the fitted line, 95 % confidence interval and prediction interval. How does this look compared to 1.A(h)?
- 1.B(i). Calculate the residuals and plot them against the predicted log-values. Also make a Q-Q-plot and a histogram. How do these look compared to 1.A(i) and 1.A(j)? Comment?

1.C Back to the original scale

Continue with Mozquizto-test *Lab 1.C: Original scale*.

- 1.C(a). Write down how the concentration of lead, Pb, develops over time, first as a function of the β -parameters in the model in 1.B, and then as a function of $a = e^{\beta_0}$ and $b = e^{\beta_1}$.
- 1.C(b). Calculate estimates of a and b together with 95 % confidence intervals.
- 1.C(c). How high was the average (or rather, median) concentration of lead in 1975, according to the estimated model?
- 1.C(d). How fast is the rate of decrease in lead concentration, according to the estimated model?
- 1.C(e). Use the model to calculate a 95 % confidence interval for the expected median lead concentration in the year `[mzq]`.
- 1.C(f). Use the model to calculate a 95 % prediction interval for an observation of the lead concentration in the year `[mzq]`.
- 1.C(g). Plot $y = \text{Pb}$ against $x = \text{year} - 1975$ together with the fitted relationship, 95 % confidence interval and prediction interval. How does this look compared to the model in 1.A(h)?
- 1.C(h). Redo the plot with $x = \text{year}$ instead. You only need to change the $x=$ in the aesthetics, everything else can be the same (except the label on the x-axis).

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LAB 2: LEAD IN MOSS — DIFFERENT REGIONS

Uses the techniques from Lecture 3 (2.A+B) and 4 (2.C)

Deadline: Friday 24 April 2026

Dedicated help sessions:

2.A+B: Monday 30 March at 8.15–10.00 and 10.15–12.00,

2.C: Wednesday 1 April at 10.15–12.00 and Thursday 2 April at 8.15–10.00.

Examination: Mozquizto Lab 2:A+B+C

Perform the tasks by copying relevant R-code from Lecture 3 and 4, rewrite and run it while answering the questions in the accompanying three Mozquizto-tests at `quizms.maths.lu.se`. These tests will also provide you with the information marked [*mzq*] below.

2 Lab 2: Problem description

This is a direct continuation of Lab 1, where we will take the locations, region, of the observations into account. The data is available in the file `Pb_all.rda`.

```
#Lab2####
library(tidyverse)
load("Data/Pb_all.rda")
summary(Pb_all)
```

2.A Does the lead level change over time?

Continue with Mozquizto-test *Lab 2.A: Change over time*

- 2.A(a). Plot Pb against year using all the data, ignoring region. Does it look like an exponential decline might work?
- 2.A(b). Fit a linear model with $y = \log(\text{Pb})$ and $x = I(\text{year} - 1975)$ using all the data.
- 2.A(c). Use a suitable test to test whether there is a significant linear relationship between $\log(\text{Pb})$ and time, i.e. $H_0: \beta_1 = 0$ against $H_1: \beta_1 \neq 0$ on significance level $\alpha = 5\%$. Report the test statistic, its degrees of freedom, the P-value, and the conclusion.
- 2.A(d). Use the model to calculate a 95 % confidence interval and a 95 % prediction interval for the lead concentration (mg/kg) in the year [*mzq*].
- 2.A(e). Calculate the residuals and plot them against the predicted log values. Also make a Q-Q-plot. Problems?
- 2.A(f). Redo the residual plots from 2.A(e) separately for each region by adding `+ facet_wrap(~region)`. Does this reveal any problems with the residuals?

2.B Add region to the model

Continue with Mozquizto test *Lab 2.B: Regional differences*.

We have measurements from five different Swedish regions and it is likely that they have different concentration levels of lead to start with. This would explain the behaviour of the residuals in 2.A(f). We will fit a model for decrease in (log) lead concentration over time assuming that the rate of deterioration is the same in all regions, but the starting levels may be different.

The `region` variable is already a factor variable. This means that R will automatically create internal dummy variables for each non-reference category. However, we might want to change the reference category.

- 2.B(a). Plot the data again, as in 2.A(a), but separately for each region by adding
`+ facet_wrap(~region)`

Does it look like the same type of relationship (exponential decline) for all regions?

- 2.B(b). Redo the plot with $\log(\text{Pb})$ on the y-axis. Does this seem like a good idea for all the regions?

- 2.B(c). Fit the linear model $\log(\text{Pb}) \sim I(\text{year} - 1975) + \text{region}$.

Which region was used as reference? Is this a good idea?

- 2.B(d). The reference category should always be a large category. Change the region variable so that `[mzq]` will be used as reference instead:

```
###2.B(d)####
Pb_all <- mutate(Pb_all, region = relevel(region, "[mzq]"))
```

Then refit the model in 2.B(c).

How high was the average concentration of lead in region `[mzq]` in 1975, according to the model? How fast is the rate of decrease in lead concentration, according to this model? Compare with 1.C(c) and 1.C(d) in Lab 1.

- 2.B(e). Use the model to calculate a 95 % confidence interval and a 95 % prediction interval for the lead concentration (mg/kg) in region `[mzq]` in the year `[mzq]`.
- 2.B(f). Estimate the ratio between the expected Pb-level in Örebro and the level in `[mzq]`, for any given year.
- 2.B(g). Örebro does not have any observations for 1975. Use the model to estimate the expected Pb-level in Örebro in 1975, together with a 95 % confidence interval.
- 2.B(h). Use the model to estimate the expected Pb-level in Örebro in the year `[mzq]`, together with a 95 % confidence interval.

2.C Tests and residuals

Continue with Mozquizto test *Lab 2.C: Tests and residuals*.

- 2.C(a). Is there a significant linear relationship between $\log(\text{Pb})$ and time, now that we have taken the different levels in the regions into account? Test $H_0: \beta_1 = 0$ against $H_1: \beta_1 \neq 0$, using a suitable test and report the test-statistic, its degrees of freedom, the P-value and the conclusion.
- 2.C(b). Are there any significant differences in the starting levels (1975) between the different regions? Test this using a suitable test, comparing the model from 2.A(b) with 2.B(d). Report the test statistic, its degrees of freedom, the P-value, and the conclusion.
- 2.C(c). Add fitted lines, confidence intervals and prediction intervals to the plot(s) in 2.B(b) and in 2.B(a). Do they seem reasonable?
- 2.C(d). Calculate the residuals and plot them against the predicted log values. Also make a Q-Q-plot. Problems? Compare with 2.A(e).
- 2.C(e). Redo the residual plots separately for each region by adding `+ facet_wrap(~region)`. Did adding the regions to the model solve the problems revealed in 2.A(f)?

In Lab 3 we will look closer at the one or two large residuals and find out if they are a problem.

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LAB 3: LEAD IN MOSS — CHECKING AND FINE TUNING

Uses the techniques from Lecture 5 (3.A) and 6 (3.B)

Deadline: Thursday 30 April 2026

Dedicated help sessions:

3.A: Tuesday 21 April at 8.15–10.00 and 10.15–12.00,

3.B: Wednesday 22 April at 10.15–12.00 and 13.15–15.00.

Examination: Mozquizto Lab 3:A+B

Perform the tasks by copying relevant R-code from Lecture 3 and 4, rewrite and run it while answering the questions in the accompanying two Mozquizto-tests at `quizms.maths.lu.se`. These tests will also provide you with the information marked [*mzq*] below.

3 Lab 3: Problem description

This is a direct continuation of Lab 2 where we validate the model and try to improve it.

```
#Lab3####
library(tidyverse)
load("Data/Pb_all.rda")
Pb_all <- mutate(Pb_all, region = relevel(region, "[mzq]"))
```

3.A Regression diagnostics

Continue with Mozquizto test *Lab 3.A: Regression diagnostics*.

- 3.A(a). Reuse the model $\log(\text{Pb}) \sim I(\text{year} - 1975)$ from Lab 2.A(b). Also fit the bad linear model $\text{Pb} \sim I(\text{year} - 1975)$ on the same, full, data.

Calculate the leverage ($\hat{v}_{ii}$) values, v_{ii} , for each model. Make one plot for each model with leverage against year and add horizontal lines at $1/n$ and $2(p+1)/n$ and make sure the y -axis includes $y = 0$.

Note: Since we have many observations each year the points will end up on top of each other. Use `geom_jitter(width = 1)` instead of `geom_point()` to add a random value to the year.

Is there any difference in the leverage between the two models? Why/why not?

Any years where the observations have a high leverage?

Which year would the leverage be at its minimum?

- 3.A(b). Now reuse the log-transformed model with all the regions using [*mzq*] as reference, from 2.B(d).

Calculate the leverage values, v_{ii} and plot them against year using different colours for the different regions, `aes(..., color = region)`. Add horizontal lines at $1/n$ and $2(p+1)/n$ and make sure the y -axis includes $y = 0$.

Any patterns of large leverage? Why are the regions ordered vertically as they are? Hint; Compare the ordering with the number of observations in each region.

- 3.A(c). For model 2.B(d), calculate the studentized residuals, r_i^* , and plot them against the linear predictor, $\hat{Y}_i$. Add horizontal lines for 0, ± 2 and ± 3 . Are there any trends in the residuals? Are there any unpleasantly large residuals?
- 3.A(d). Redo the plot separately for each region using `+ facet_wrap(~region)`. Are there any trends in the residuals in any of the regions? What might that indicate?
- 3.A(e). Plot $\sqrt{|r_i^*|}$ against the linear predictor, $\hat{Y}_i$, separately for each region. Make sure the y -axis contains $y = 0$ and add horizontal lines for $\sqrt{\lambda_{0.25}}$, $\sqrt{2}$ and $\sqrt{3}$. Are there any unpleasant trends in the residual variability in any of the regions?
- 3.A(f). Calculate Cook's Distance and plot them against year, separately for each region. Add horizontal lines at $F_{0.5, p+1, n-(p+1)}$ and $4/n$ and make sure the y -axis includes $y = 0$.
Are there any observations with a really large Cook's distance? If not, redo the plot without the $F_{0.5, p+1, n-(p+1)}$ -reference line.
Are there any observations that have had a substantially larger influence on the estimates than the rest? Were they also observations with high leverage? Were they also observations with large residuals?
- 3.A(g). Calculate DFBETAS and save the ones for the time variable. Plot them against time, separately for each region. Did the influential points in 3.A(f) in Örebro have a large influence on the estimate of the rate of decline? Did the one in Västra Götaland?
- 3.A(h). Identify the influential point in Västra Götaland, redo the plot in 2.B(a) and highlight the point in red. Does it seem logical that it had a large influence on the estimated rate of decline?

3.B Model selection

Continue with Mozquizto test *Lab 3.B: Model selection*.

- 3.B(a). Fit a model with interaction between time and region,

$$\log(\text{Pb}) \sim \text{I}(\text{year} - 1975) * \text{region}$$
 and use a suitable test to determine whether this gives a significant improvement. Is the rate of decline the same in all regions? Also redo the residual plot in 3.A(d). Has anything happened to the trends?
- 3.B(b). Calculate R^2 , R^2_{adj} , AIC and BIC for all three models, 2.A(b) (only time), 2.B(d) (time and region) and 3.B(a) (interaction). Do the measures agree on which model is best? Worst?
 For the model that is best according to AIC, how much of the variability in log-lead concentration does it explain?